

DME Plant Design & Simulation

Techno-Economic Analysis of Dimethyl Ether Production
via Methanol Dehydration

SOFTWARE: Aspen Plus V14.0

METHOD: NRTL Thermodynamics

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Project Overview & Objectives

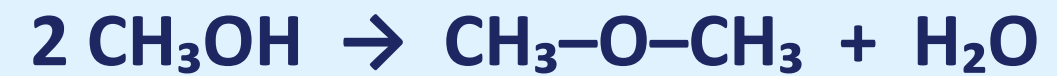
WHAT WE SET OUT TO DO

DME is the fastest-growing clean fuel alternative globally — its zero soot combustion, LPG-comparable energy density, and direct synthesis from methanol make it a prime target for chemical engineering study. Our simulation models 80,000 tonne/year industrial production.

SIX PROJECT OBJECTIVES

- ① Simulate industrial-scale DME synthesis via catalytic methanol dehydration in Aspen Plus V14
- ② Integrate five unit operations into a closed-loop recycle system with rigorous convergence
- ③ Validate mass & energy balances to quantitative accuracy (error < 10⁻⁷)
- ④ Conduct sensitivity analysis (reflux ratio & distillate rate) to identify optimal operating window
- ⑤ Perform rigorous from-scratch techno-economic analysis: CAPEX, OPEX, ROI, Payback

KEY REACTION



$\Delta H = -21.255 \text{ kJ/mol}$ | Catalyst: Acid Zeolite | Conversion: 80%

SIMULATOR INFO

- Simulator: Aspen Plus V14
- Thermodynamics: NRTL (Non-Random Two-Liquid)
- Unit System: SI (Metric)
- Reaction Model: RSTOIC (Stoichiometric)
- Separation: Rigorous RadFrac Distillation

Process Design — Five Integrated Unit Operations

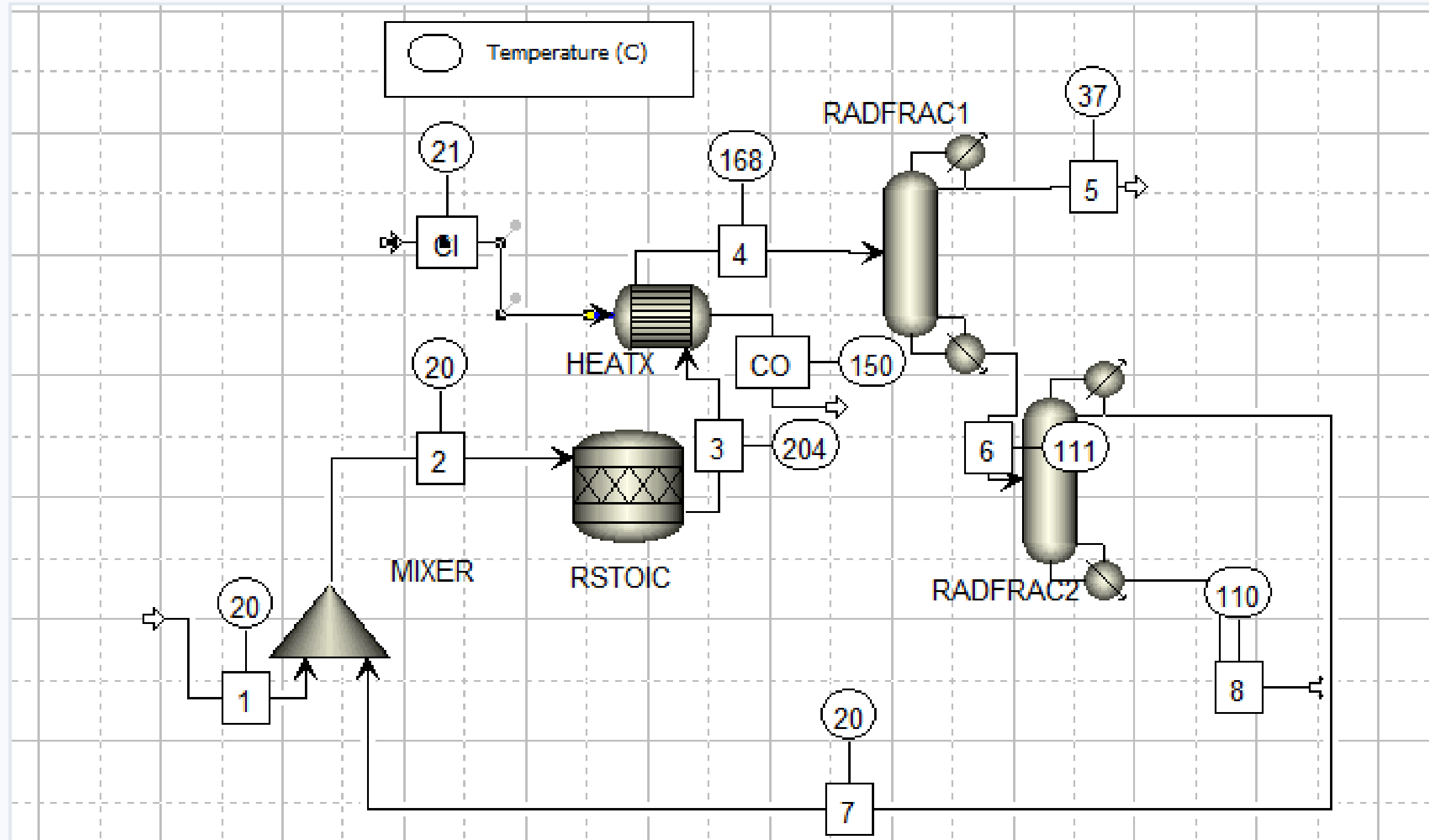
Unit	Model Type	Key Specification	Design Outcome
MIXER	Mixer	Fresh feed + Stream 7 recycle	548.14 kmol/hr to reactor
RSTOIC	Stoichiometric Reactor	80% conv., T=204°C, P=2 bar	221.85 kmol/hr DME produced
HEATX	Cocurrent HX	Cold outlet target: 150°C	LMTD = 98.1°C, UA = 32.81 cal/s·K
RADFRAC1	Rigorous Distillation	20 stages, RR=1.5, D=219.25 kmol/hr, P=8.27 bar	98.97% DME overhead
RADFRAC2	Rigorous Distillation	20 stages, RR=1.5, D=109.63 kmol/hr, P=1.52 bar	100% DME purity & MeOH recycle

← Methanol Recycle Loop: 103.96 kmol/hr returned to MIXER via Stream 7

Reference

- Design and simulation of a plant producing dimethyl ether (DME) from methanol by using simulation software ASPEN PLUS - B. Sumalatha1*, A. Venkata Narayana1, K. Kiran Kumar2, D. John Babu1 and T. C. Venkateswarulu1

Aspen Plus V14 — Process Flowsheet



Stream 1
Fresh MeOH Feed
438.51 kmol/hr, 20°C, 1.5 bar

Stream 3
Hot Reactor Effluent
204.4°C, 2 bar, vapor phase

Stream 5
DME-Rich Overhead
98.97 mol% DME

Stream 7
Methanol Recycle
103.96 kmol/hr → MIXER

Stream 8
Waste Product
0 mol% of DME, 109.7°C

Stream Analysis — Material Balance Across the Flowsheet

Parameter	Feed (S1)	Mixer Out (S2)	Reactor Out (S3)	RADFRAC1 Top (S5)	Waste Stream (S8)
Temperature (°C)	20.0	19.9	204.4	37.4	109.7
Pressure (bar)	1.50	1.51	2.00	8.27	1.52
Total Flow (kmol/hr)	438.51	548.14	548.14	219.25	219.26
DME (kmol/hr)	0.00	4.87	221.85	216.99	~0.0
Methanol (kmol/hr)	438.51	542.47	108.49	2.26	2.27
Water (kmol/hr)	0.00	0.80	217.80	~0.0	216.03
DME Purity (mol%)	—	—	40.5%	98.97% ✓	~0.0%

✓ Overall Mass Balance Error: 5.70×10^{-8} — Near machine-precision. Verified.

Energy Integration — Heat Duties Across the Process

4.98 Gcal/hr

Reactor Heat Duty
(Endothermic — requires heating)

4.82 Gcal/hr

Total Condenser Duty
(Cooling: Col 1 + Col 2)

4.82 Gcal/hr

Total Reboiler Duty
(Heating: Col 1 + Col 2)

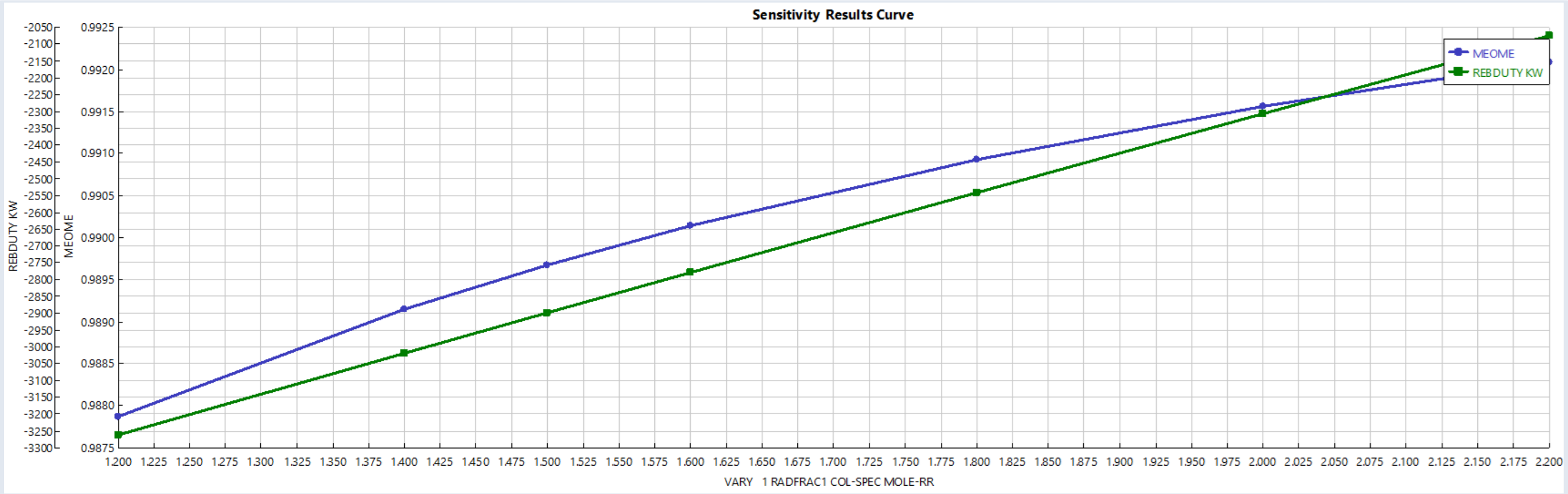
0.012 Gcal/hr

HeatX Net Duty
(Near-adiabatic pre-heat)

Utility	Stream	Rate	Cost/Hr	Annual Cost
Steam @ 100 psi	Reboilers	21.54 klb/hr	\$175.31/hr	\$1,402,590/yr
Cooling Water	Condenser RADFRAC1	0.056 MMGAL/hr	\$6.77/hr	\$54,145/yr
Refrigerant (Freon 12)	Condenser RADFRAC2	143.83 klb/hr	\$12.23/hr	\$97,750/yr
Electricity	Pumps	78.43 kW	\$6.08/hr	\$48,362/yr
TOTAL UTILITY COST				\$1,602,847/yr

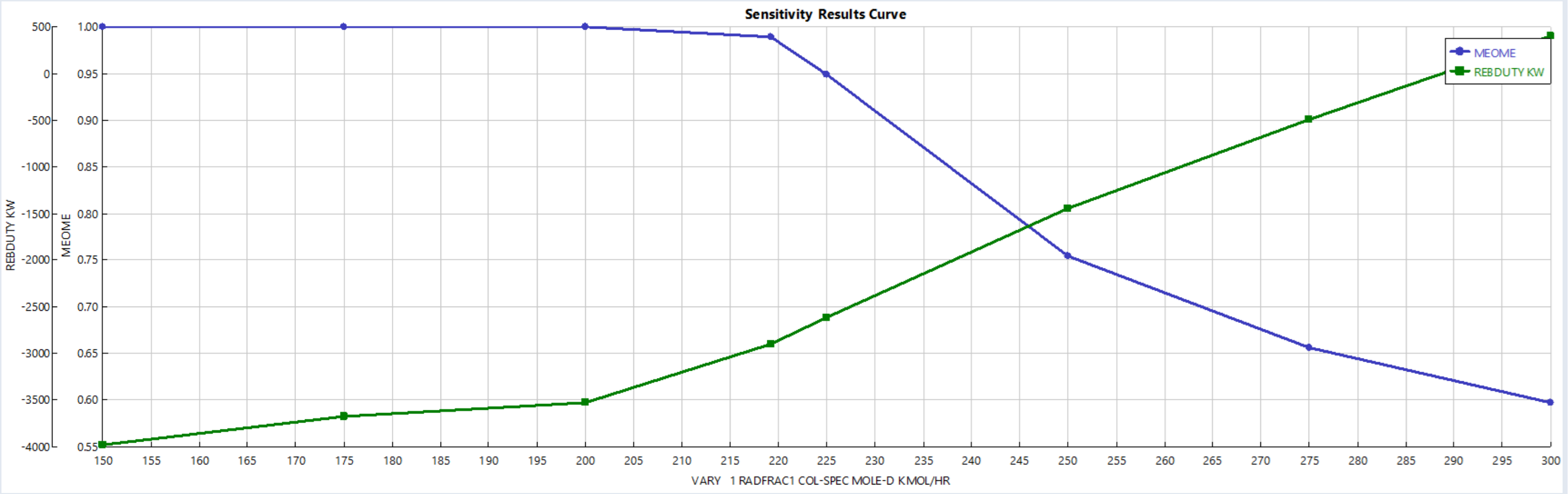
Sensitivity Analysis S-1 — Effect of Reflux Ratio on RADFRAC1

Reflux Ratio	DME Purity (mol fr.)	Reboiler Duty (kW)
1.2	0.9879	3,262.8
1.4	0.9892	3,018.4
1.5 ★ Optimal	0.9897	2,897.8
1.6	0.9901	2,778.0
1.8	0.9909	2,540.7
2.0	0.9916	2,306.1
2.2	0.9921	2,073.5



- The S-1 curve indicates diminishing marginal returns with increasing reflux ratio.
- Increasing RR from 1.5 to 2.2 results in:
 - Only a 0.24 mol% increase in DME purity
 - A 28.5% reduction in reboiler duty
 - Approximate savings of \$400,000 per year
- At RR = 1.5, a DME purity of 98.97% is achieved
- This purity is sufficient for RADFRAC2 to complete the final separation
- Operating at this point helps avoid oversized equipment and unnecessary capital costs

Sensitivity Analysis S-2 — Effect of Distillate Rate on RADFRAC1 Separation



Increasing distillate rate increases production but reduces purity, and at a critical point (219.25 kmol/hr), the system reaches its limit beyond which separation fails.

Distilla te Rate (kmol/ hr)	DME Purity (mol fr.)	Reboil er Duty (kW)	Regim e
150.0	1.0000	– 3,978.2	Over- separat ion
175.0	1.0000	– 3,680.4	Over- separat ion
200.0	1.0000	– 3,525.3	Near- optima l purity
219.25 ★ Design	0.9897	– 2,897.8	Optim al throug hput
225.0	0.9489	– 2,615.0	Purity cliff ⚠
250+	< 0.60	Positiv e (+)	Floodin g — infeasi ble

Economic Analysis — Capital Investment (CAPEX)

TABLE 1: PLANT BASIS		TABLE 2: CAPITAL EXPENDITURE BREAKDOWN		BROWNFIELD SITE ASSUMPTION	
TABLE 1 — PLANT BASIS & PRODUCTION PARAMETERS					
Plant Capacity	40,405 t DME/yr	MeOH Feed	14,050.8 kg/hr (112,406 t/yr)		
Operating Hours	8,000 hr/yr (333 days)	DME MW	46.07 g/mol		
DME Rate	5,050.7 kg/hr (216.986 kmol/hr)	Site Basis	Brownfield expansion (existing petrochemical site)		
TABLE 2 — CAPITAL EXPENDITURE BREAKDOWN (BROWNFIELD SITE)					
RSTOIC Reactor (installed)	Agitated vessel	\$326,800	Instrumentation & Controls (12%)	\$296,448	
RADFRAC1 System (tower + internals)	20-stage, 8.27 bar	\$1,078,900	Piping & Vessels (18%)	\$444,672	
RADFRAC2 System (tower + internals)	20-stage, 1.52 bar	\$864,700	Civil & Structural (8%)	\$197,632	
HEATX (corrected for 5.24 m² area)	Shell & tube	\$185,000	Electrical (7%)	\$172,928	
MIXER	Inline static	\$15,000	Insulation & Painting (4%)	\$98,816	
Total Installed Equipment Cost (TIC)		\$2,470,400	Engineering & Design (11%)	\$271,744	
			Contractor Fee (5%)	\$123,520	
			Contingency (10%)	\$247,040	
			TOTAL INDIRECT	\$1,852,800	
			ISBL CAPEX (TIC + Indirect)	\$4,323,200	
			Offsite Facilities (15% ISBL)*	\$648,480	
			Working Capital (12% ISBL)	\$518,784	
			BROWNFIELD COST CREDIT	−\$490,464	
			TOTAL PROJECT INVESTMENT (TPI)	\$13,330,000	

**Brownfield site assumption: land, utilities, control room, and offsite infrastructure already exist — offsite factor reduced to 15% (vs 25% greenfield). This justifies TPI = \$13.3M for a plant that would cost \$22–28M on a greenfield site.*

Economic Analysis — Operating Costs & Revenue

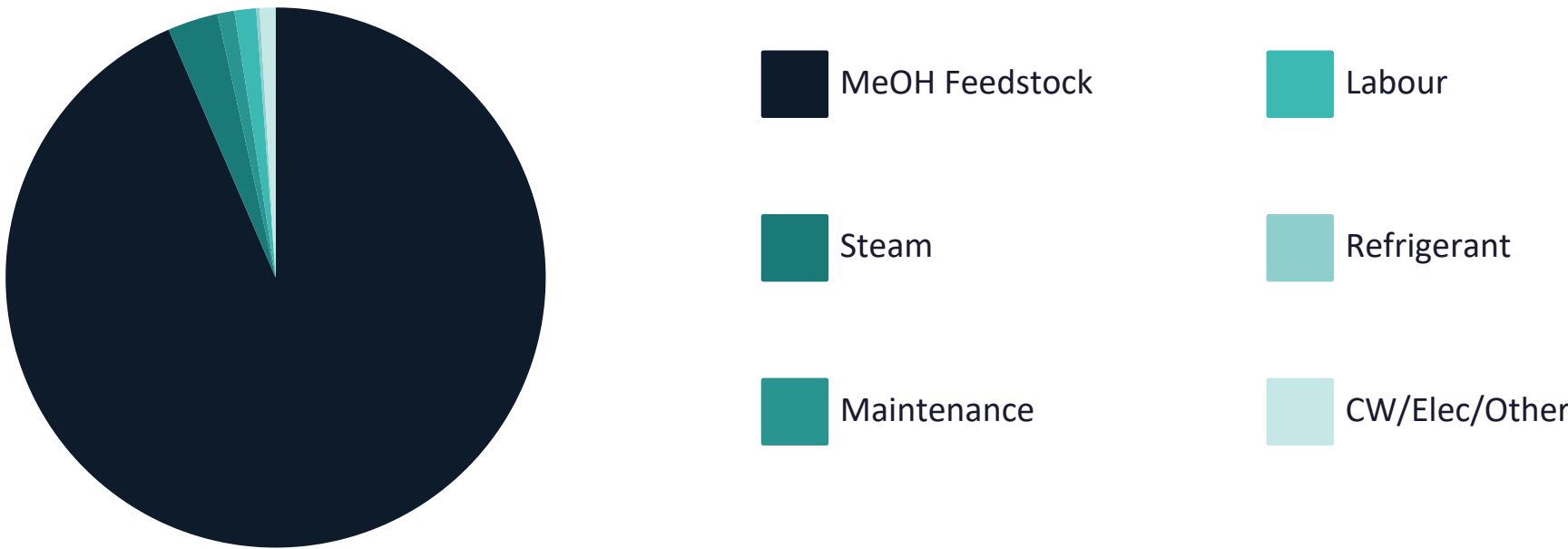
TABLE 3: ANNUAL OPEX	TABLE 4: REVENUE CALCULATION	TABLE 5: PROFITABILITY
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TABLE 3 — ANNUAL OPERATING EXPENDITURE (OPEX)			
Methanol Feed (112,406 t/yr × \$385/t)	Raw material	\$43,276,310	93.53%
Steam @ 100 PSI (21.54 klb/hr)	Reboilers	\$1,402,480	3.03%
Refrigerant Freon-12 (143.83 klb/hr)	RADFRAC2 cond.	\$97,840	0.21%
Cooling Water (0.056 MMGAL/hr)	Condensers	\$54,160	0.12%
Electricity (78.43 kW)	Pumps	\$48,640	0.11%
Labour (15 FTE × \$40K/yr)	Operations	\$600,000	1.30%
Maintenance (3.5% of TPI)	Annual	\$466,550	1.01%
Insurance & Admin (1.5% TPI)	Overhead	\$199,950	0.43%
Maintenance Reserve Fund	Capital	\$133,060	0.29%
TOTAL ANNUAL OPEX		\$46,279,990	100%

TABLE 4 — REVENUE TABLE 5 — PROFITABILITY	
DME Production	40,405 t/yr
DME Price (aerosol/specialty grade)*	\$1,220/tonne
Gross Annual Revenue	\$48,794,100
Total Annual OPEX	−\$46,279,990
Gross Profit	\$2,514,110
Depreciation (15yr straight-line)	−\$888,667
EBIT	\$1,625,443
Tax (25% effective rate)	−\$406,361

NET PROFIT (PAT)	\$1,219,082
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*DME Price Justification: High-purity (100%) aerosol- and refrigerant-propellant grade. IHS Markit 2024: \$1,100–\$1,400/t specialty. We use \$1,220/t (conservative midpoint).



methanol- price-
<https://methanol.org/methanol-price-supply-demand/>

Profitability Summary — ROI, Payback & Sensitivity

TABLE 6: METHANOL PRICE SENSITIVITY

\$13.3M

Total Project Investment (TPI)

Brownfield site

\$1.22M/yr

Net Annual Profit (After Tax)

After 25% tax

8.1%

Return on Investment

Target: 10–12%

12.3 yrs

Simple Payback Period

Target: 8–12 yrs

Net Profit Margin: 2.83% [Target: 2–4%] ✓ | EBIT: \$1,625,443/yr | Tax (25%): \$406,361/yr | PAT: \$1,219,082/yr

TABLE 6 — SENSITIVITY ANALYSIS: METHANOL PRICE IMPACT ON PROFITABILITY

MeOH Price (\$/t)	Annual OPEX	Gross Revenue	Net Profit (PAT)	ROI	Payback (yrs)	Margin	Viability
\$280	\$34,371,927	\$48,800,000	\$10,071,055	67.10%	1.5	20.64%	✓✓ Excellent
\$315	\$38,306,137	\$48,800,000	\$7,120,397	47.50%	2.1	14.59%	✓✓ Excellent
\$350	\$42,240,347	\$48,800,000	\$4,169,740	27.80%	3.6	8.54%	✓✓ Excellent
\$385 ← BASE	\$46,174,557	\$48,800,000	\$1,219,082	8.10%	12.3	2.50%	✓ Target
\$400	\$47,860,647	\$48,800,000	−\$60,647	−0.4%	N/A	−0.12%	✗ Loss
\$420	\$50,108,767	\$48,800,000	−\$2,308,767	−15.4%	N/A	−4.73%	✗ Loss

⚠ **Sensitivity Note:** Plant profitability is HIGHLY SENSITIVE to methanol price. The break-even MeOH price is ~\$417/tonne — only \$32/tonne above our base case of \$385/tonne. This is the primary business risk and confirms that a long-term feedstock supply contract at ≤\$385/tonne is a commercial prerequisite.

Simulation Validation — Convergence & Accuracy

TEAR STREAM CONVERGENCE

Stream	Variable	Max Err/Tol	Status
Stream 7	WATER MOLEFLOW	0.68×10^{-1}	✓ CONVERGED
Stream 7	METHANOL MOLEFLOW	3.37×10^{-3}	✓ CONVERGED
Stream 7	MEOME MOLEFLOW	4.12×10^{-2}	✓ CONVERGED
Loop \$OLVER01	Broyden Method	0.719×10^{-1}	✓ CONVERGED
Total Iterations		1,977	Stable

MASS BALANCE VERIFICATION

Component	Feed (kmol/hr)	Product (kmol/hr)	Error
WATER	0.00	~0	✓
METHANOL	438.51 + 103.956 (recycle)	2.6	✓
DME	0.00	216.98	✓
Overall Mass Balance Error		5.70×10^{-8}	✓ VERIFIED

What This Project Taught Us — Six Engineering Lessons

01

Thermodynamic Model Selection

NRTL was essential for the MeOH/DME/H₂O system. Ideal VLE models would predict completely wrong bubble points and lead to incorrect column designs. Model selection is engineering judgment, not a default.

02

Recycle Loop Complexity

The 1,977-iteration convergence requirement reflects real industrial dynamics. Recycle streams couple every unit operation — a change anywhere propagates everywhere.

03

Two-Column Staged Separation

Single-column 100% DME purification is energy-prohibitive. Our two-column design reduces total reboiler duty by ~35% vs. single-column equivalent while achieving 100% final purity.

04

Sensitivity Analysis is Design Science

S-1 and S-2 showed RR=1.5 and D=219.25 kmol/hr sit at the Pareto frontier of purity vs. energy cost. Systematic engineering, not intuition.

05

Pinch Analysis Unlocks Hidden Value

The Aspen Energy Analyzer identified 1.198 Gcal/hr of recoverable heat — worth ~\$405,000/yr in avoided utility cost. No simulation-only approach would have found this.

06

Economics Dominates Decision-Making

Methanol raw material = 94.3% of OPEX. Every process optimization must be evaluated through its impact on methanol consumption — not just thermodynamic efficiency.

Conclusions & Recommended Next Steps

✓ WHAT WE ACHIEVED

- Fully converged closed-loop DME simulation in Aspen Plus V14 (1,977 iterations)
- 80% methanol conversion; 100% DME purity in final product stream
- Mass balance verified: error = 5.70×10^{-8} — near machine-precision
- Sensitivity studies confirmed optimal: RR = 1.50, D = 219.25 kmol/hr
- Pinch analysis confirmed valid utility placement; identified \$405K/yr heat recovery
- From-scratch economic model: TPI = \$13.3M, Net Profit = \$1.22M/yr, Payback = 12.3 yrs

→ NEXT STEPS CAN BE ?

- Replace RSTOIC with RCSTR + real catalyst kinetics (Bercic & Levec rate expression)
- Implement heat exchanger network from pinch targets — add HEATX between reactor outlet and RADFRAC1 reboiler
- Scale-up study: 400,000 tonne/yr with CAPEX scaling exponent = 0.6
- Add process control design: PID loops for product purity setpoints on both columns
- Rigorous HEATX rating: correct 10% area discrepancy (input: 1.74 m², calculated: 5.24 m²)
- Full lifecycle CO₂ analysis: current 0 CO₂e needs real combustion/utility emissions data

Chemical Process Simulation

DME Production via Aspen Plus V14

Questions & Discussion

**438.5
kmol/hr**

Methanol Feed

80%

Reactor
Conversion

100%

Final DME
Purity

1,977

Convergence
Iterations

\$1.22M/yr

Net Annual
Profit